Post: Think-aloud User Study on GenAI for Software Engineering Tasks

This post-survey takes approximately 3-5 minutes to complete. Your responses will help us understand your usage patterns, the effectiveness, perceived learning outcomes, workload impacts, and general opinions of GenAI in solving SE tasks, and knowledge assessment of the software engineering (SE) concepts.
Please let the researcher know if you have any questions.

* Indicates required question

1. Email *

Usage Patterns and Self-Rated Effectiveness

2. For each task, which modalities did you use? (Select all that apply per task) *

Check all that apply.

	Text prompts	Media upload	Audio/voice conversation	Video/screen sharing	N/A
Requirement Review	☐	☐	☐	☐	☐
Usabiltiy Review	☐	☐	☐	☐	☐
Feature Implementation (Design-to-Code)	☐	☐	☐	☐	☐

3. *Please rate your agreement with the following statement for each interaction mode:* * *"This [interaction mode] helped me complete the **Requirement Review** task effectively."*

Mark only one oval per row.

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	N/A
Text prompts	◯	◯	◯	◯	◯	◯
Media upload	◯	◯	◯	◯	◯	◯
Audio/voice conversation	◯	◯	◯	◯	◯	◯
Video/screen sharing	◯	◯	◯	◯	◯	◯

4. *Please rate your agreement with the following statement for each interaction mode:* * *"This [interaction mode] helped me complete the **Usability Review** task effectively."*

Mark only one oval per row.

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	N/A
Text prompts	◯	◯	◯	◯	◯	◯
Media upload	◯	◯	◯	◯	◯	◯
Audio/voice conversation	◯	◯	◯	◯	◯	◯
Video/screen sharing	◯	◯	◯	◯	◯	◯

5. *Please rate your agreement with the following statement for each interaction mode:* * *"This [interaction mode] helped me complete the **Feature Implementation** task effectively."*

Mark only one oval per row.

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	N/A
Text prompts						
Media upload						
Audio/voice conversation						
Video/screen sharing						

Workload Measurement & Learning Effects

6. How complex was each task itself, regardless of the AI tool? *

Mark only one oval per row.

	Very low	Low	Moderate	High	Very High
Requirement Review					
Usabiltiy Review					
Feature Implementation (Design-to-Code)					

7. How much extra effort was required to interact with each modality of the GenAI tool, beyond the task itself? *

Mark only one oval per row.

	Very Low	Low	Moderate	High	Very high	N/A
Text prompts	◯	◯	◯	◯	◯	◯
Media Upload	◯	◯	◯	◯	◯	◯
Audio/Voice conversation	◯	◯	◯	◯	◯	◯
Video/screen sharing	◯	◯	◯	◯	◯	◯

8. How much extra effort did it take to switch across modalities during the study? *

Mark only one oval.

	1	2	3	4	5	
Very	◯	◯	◯	◯	◯	Very high extra effort

9. To what extent do you understand the tasks with the help of the GenAI tool? *

Mark only one oval per row.

	Not at all	Slightly	Moderately	Very well	Completely	N/A
Requirement Review						
Usabiltiy Review						
Feature Implementation (Design-to-Code)						

10. Please rate the following statement for each concept: *
"After using the Multimodal GenAI tool during the study, my understanding of [concept] has increased compared to before."

Mark only one oval per row.

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	N/A
Software Requirement Specification						
Functional Requirements						
Use Case						
User Story						
Nielsen's Usability Heuristics						

Overall Experience & Opinions

11. Compared to text-only GenAI, multimodal interaction (media upload/voice input/screen-sharing) felt more effective in solving **Requirement Review** task. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

12. Compared to text-only GenAI, multimodal interaction (media upload/voice input/screen-sharing) felt more effective in solving **Usability Review** task. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

13. Compared to text-only GenAI, multimodal interaction (media upload/voice input/screen-sharing) felt more effective in solving **Feature Implementation** task. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

14. For which SE task did you find the multimodal GenAI features most helpful? *

Mark only one oval.

- ◯ Requirement Review
- ◯ Usability Review
- ◯ Feature Implementation (Design-to-Code)

Skip to question 15

Knowledge Assessment of the Software Engineering Concepts

15. What is the primary purpose of a Software Requirements Specification (SRS) document? *

Mark only one oval.

- To define the visual branding and marketing materials for launch
- To define what the system must do and guide its design
- To act as a permanent contract that can never be revised after signing
- To document defects found once the product is live

16. Which of the following would typically belong inside an SRS? *

Mark only one oval.

- A list of all Git commit messages from the sprint
- The team's daily stand-up notes from every week's sprint
- A functional requirement describing what the system must do
- The full source code written for the unit tests

17. What is the primary focus of a use case in software engineering? *

Mark only one oval.

- The exact syntax of the programming language used to build the system
- The estimated cost and timeline of the project
- A goal-oriented interaction between a user and the system
- The physical servers and network topology the system runs on

18. For an ATM's 'Withdraw Cash' use case, which of the following would be listed as a precondition? *

Mark only one oval.

- The customer's account balance is updated after withdrawal
- The customer receives their cash from the machine
- The bank designates the customer as the primary actor
- The customer's card has been inserted and PIN verified

19. What is the main purpose of a user story in software development? *

Mark only one oval.

- To specify the exact user interface layout and pixel measurements
- To provide a short, user-focused description of a feature
- To serve as a binding legal agreement with the client
- To document every edge case and exception before development starts

20. A user story reads: "As a user, I want to reset my password, so that I can regain access to my account." Which of the following is an example of acceptance criteria for this story? *

Mark only one oval.

- "As a user, I want to reset my password, so that I can regain access to my account."
- "Given a valid email, when the user requests a reset link, then an email is sent within 1 minute.
- "The database schema shall include a table called password_reset_tokens for storing reset requests."
- "The team has estimated that this story is worth approximately 3 story points."

21. Which of the following is an example of a functional requirement? *

Mark only one oval.

- The system must encrypt all sensitive data while at rest
- The system should load all pages within 500 milliseconds
- The system shall allow users to reset their password via email
- The system must maintain 99.99% uptime during business hours

22. Which pairing correctly matches each requirement to its type? *

Mark only one oval.

- "Checkout page must load in under 1 second" is functional; "Users can apply a discount code" is non-functional
- "Users can apply a discount code" is functional; "Checkout page must load in under 1 second" is non-functional
- Both statements are functional requirements
- Both statements are non-functional requirements

23. What is the purpose of usability heuristics (e.g., Nielsen's 10)?

Mark only one oval.

- To calculate the return on investment for a redesign
- To provide general principles that help evaluate interface usability
- To define the exact color palette a product must use
- To specify the programming framework a team should adopt

24. A website shows a plain spinning loader with no message or progress indicator during a 30-second file upload. Which usability heuristic is being violated? *

Mark only one oval.

- Aesthetic and minimalist design
- Consistency and standards
- Visibility of system status
- Recognition rather than recall

Knowledge Assessment of the Software Engineering Concepts

25. What is the primary purpose of a Software Requirements Specification (SRS) document? *

Mark only one oval.

- To define the visual branding and marketing materials for launch
- To establish a shared, agreed description of what the system must do, guiding design and development
- To act as a permanent contract that can never be revised after signing
- To document defects found once the product is live

26. Which of the following would typically belong inside an SRS? *

Mark only one oval.

- A list of all Git commit messages from the sprint
- The team's daily stand-up notes
- A functional requirement describing what the system must do
- The source code for the unit tests

27. What is the primary focus of a use case in software engineering? *

Mark only one oval.

- The exact syntax of the programming language used to build the system
- The estimated cost and timeline of the project
- A goal-oriented interaction between a user and the system, described from the user's viewpoint
- The physical servers and network topology the system runs on

28. For an ATM's 'Withdraw Cash' use case, which of the following would be listed as a precondition? *

Mark only one oval.

- The customer's account balance is updated to reflect the withdrawal
- The customer receives their cash from the machine
- The bank designates the customer as the primary actor
- The customer's card has been successfully inserted and their PIN verified

29. What is the main purpose of a user story in software development? *

Mark only one oval.

- To specify the exact UI layout and pixel measurements
- To provide a short, user-focused description of a feature
- To serve as a binding legal agreement with the client
- To document every edge case and exception before development starts

30. A user story reads: "As a user, I want to reset my password, so that I can regain access to my account." Which of the following is an example of acceptance criteria for this story? *

Mark only one oval.

- "As a user, I want to reset my password, so that I can regain access to my account."
- "Given a valid email address, when the user requests a reset link, then an email containing the link is sent within 1 minute."
- "The database shall include a password_reset_tokens table."
- "The team estimates this story at 3 story points."

31. Which of the following is an example of a functional requirement? *

Mark only one oval.

- The system must encrypt all data at rest
- The system should load pages within 500ms
- The system shall allow users to reset their password via email
- The system must maintain 99.99% uptime

32. Which pairing correctly matches each requirement to its type? *

Mark only one oval.

- "Checkout page must load in under 1 second" is functional; "Users can apply a discount code" is non-functional
- "Users can apply a discount code" is functional; "Checkout page must load in under 1 second" is non-functional
- Both statements are functional requirements
- Both statements are non-functional requirements

33. What is the purpose of usability heuristics (e.g., Nielsen's 10)?

Mark only one oval.

- To calculate the return on investment for a redesign
- To provide broad, general principles that help evaluate and guide interface usability
- To define the exact color palette a product must use
- To specify the programming framework a team should adopt

34. A website shows a plain spinning loader with no message or progress indicator during a 30-second file upload. Which usability heuristic is being violated? *

Mark only one oval.

- Aesthetic and minimalist design
- Consistency and standards
- Visibility of system status
- Recognition rather than recall

Knowledge Assessment of the Software Engineering Concepts

35. What is the primary purpose of a Software Requirements Specification (SRS) document? *

Mark only one oval.

- To document the internal code architecture and algorithms
- To outline the project's marketing and sales plan
- To serve as the agreed baseline before development begins
- To log bugs discovered during testing

36. Per IEEE guidance, which of these is considered a quality attribute of a good SRS? *

Mark only one oval.

- Unverifiable, to avoid limiting the design
- Written only in code, not natural language
- Ambiguous, so it can be interpreted flexibly
- Complete and consistent, with no conflicts

37. What is the primary focus of a use case in software engineering? *

Mark only one oval.

- The programming language used to build the system
- What the system does from the user's perspective
- The specific hardware the system runs on
- How the internal code is structured

38. In a use case, what does an 'actor' represent? *

Mark only one oval.

- A specific function in the source code
- A step in the testing checklist
- A table in the system's database
- A person or system interacting externally

39. What is the main purpose of a user story in software development? *

Mark only one oval.

- To capture a short description of desired functionality of a system
- To serve as a complete, final technical specification
- To describe the CI/CD deployment pipeline
- To formally document the system's non-functional requirements

40. Which of these is written in a typical user story format? *

Mark only one oval.

- "Section 4.2: Password Reset Functionality Specification Document"
- "Use Case: Reset Password. Actor: Customer. Preconditions: account exists..."
- "The system shall process 1,000 transactions per second."
- "As a customer, I want to reset my password, so that I can regain access to my account."

41. Which of the following is an example of a functional requirement? *

Mark only one oval.

- The system must be available 99.9% of the time
- The system shall allow login with username and password
- The system must respond to any request within 2 seconds
- The system must comply with GDPR data-retention rules

42. What do functional requirements primarily describe? *

Mark only one oval.

- The specific services and behaviors the system provides
- The budget and timeline for development
- The hardware needed to run the system
- Quality attributes like speed, security, or reliability

43. What is the purpose of usability heuristics (e.g., Nielsen's 10)? *

Mark only one oval.

- To specify exact pixel dimensions for every UI element
- To provide broad rules for evaluating usability
- To calculate the total cost of development
- To define the database schema for an application

44. Which of the following is one of Nielsen's 10 usability heuristics? *

Mark only one oval.

- Continuous integration
- Object-oriented encapsulation
- Visibility of system status
- Database normalization

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